Net Revenue Management- A FMCG Case Study

Rohith Vaddepally

FMCG: FMCG stands for Fast Moving Consumer Goods. Colgate, Pepsi, Coca-Cola, Nivea, Nutella, Oreo, etc., are some FMCG brands. Good examples are food, beverages, personal care items or household cleaning products. These products have a relatively short lifespan. These items are typically quite cheap, widely available, and in high demand. Suppliers sell them in huge volumes at relatively low margins.

Case Study: This case study analyses the application of Net Revenue Management (NRM) principles to drive sustainable and profitable growth for Healthmax, a shampoo business company. I acted as the Category Manager for Healthmax, with a focus solely on increasing sales volume, improving revenue quality by analysing Key Performance Indicators, applying NRM techniques such as Brand Portfolio Pricing & Mix Management, and optimising net revenue through Pack Price Architecture, new pricing strategy, and forecasting future revenue.

This case study is split into three sections:

1. Fast Moving Consumer Goods
2. Net Revenue Management
3. Optimising Net Revenue

This structured approach demonstrates how NRM can transform analytical insights into actionable strategies that support long-term, profitable business growth for Healthmax.

Dataset: Initially, the project has 4 datasets to work on. They are

1. Dataset_shampoo
2. Internal_sales_data
3. New_product_launch
4. Promotion_analysis

	A	B	C	D	E	F	G	H	I	
1	Category	Subcategory	Supplier	Brand	Region	Year	Month	Units Month	Values Month	Dat
2	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	1	11560	$ 62,800	01/
3	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	2	10750	$ 58,360	01/
4	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	3	11020	$ 59,840	01/
5	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	4	11310	$ 61,400	01/
6	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	5	13780	$ 74,870	01/
7	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	6	13260	$ 72,050	01/
8	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	7	11710	$ 63,620	01/
9	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	8	12800	$ 69,500	01/
10	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	9	9140	$ 49,640	01/
11	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	10	11690	$ 63,460	01/
12	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	11	9520	$ 51,710	01/
13	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2018	12	13110	$ 71,200	01/
14	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	1	9280	$ 55,560	01/
15	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	2	11380	$ 68,140	01/
16	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	3	10990	$ 65,840	01/
17	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	4	11920	$ 71,430	01/
18	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	5	10080	$ 60,330	01/
19	Shampoo	Anti-dandruff	Apex Trading Co.	RedRose	Center	2019	6	9190	$ 55,050	01/

Fast Moving Consumer Goods:

1. **Number of brands for each supplier**: Created a pivot table for supplier name and brand count. Healthmax supplier has a total of 378 brands associated with it.

Supplier Name	Brands
Apex Trading Co.	738
AquaSolutions	189
Badoa	183
Blue Sky Enterprises	155
Elite Cosmetics	126
FreshCo Industries	378
Glitterati	189
GreenLeaf Distributors	566
HealthMax	378
Ivory Beaty	166
Luminious Skincare	173
Nature's Harvest	189
Opulent Beauty Co.	187
Parilux	182
Sparkle & Shine	306
Superior Supplies Co.	138
Sustaina	189
Grand Total	**4432**

2. **Healthmax Growth**: Its time to analyse the Healthmax supplier growth from 2019-2022. Ignoring 2023 as the data is only available until March, and it is not suggested to compare it with the previous year, 2022. Healthmax has two brands associated with it, **Shinez** & **Starbust**. The pivot table below calculates the value growth from 2018 to 2022, calculating the difference in value performance from the previous year.

Supplier	HealthMax				
Sum of Values Month	**Column Labels**				
Row Labels	**2018**	**2019**	**2020**	**2021**	**2022**
Shinez		-1.89%	7.66%	0.90%	6.74%
Starbust		-6.02%	1.77%	1.24%	-0.01%
Grand Total		**-4.64%**	**3.79%**	**1.12%**	**2.38%**

Year **2020** performed well in the value sales from its previous year, with **3.79%,** and the brand Shinez was always performing better than Starbust.

3. **Year-to-date KPI (Brand: Redrose, Region: Center)**:

```
=SUMIFS(I:I,D:D,D2,E:E,E2,F:F,F2,G:G,"<="&G2)
```

A new column is calculated for the year-to-date value totals, and units totals with the above formula, which gives the total value until that current date. As a Category manager, I created a pivot table for the YTD value for the brand Redrose and for the central region from 2018 to 2023.

Brand	RedRose					
Region	Center					
Sum of Values YTD	**Column Year**					
Row Month	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**
1	$ 62,800	$ 55,560	$ 48,950	$ 53,620	$ 39,930	$ 29,790
2	$ 121,160	$123,700	$ 92,340	$ 95,470	$ 76,270	$ 68,760
3	$ 181,000	$189,540	$155,430	$144,020	$117,300	$108,030
4	$ 242,400	$260,970	$208,940	$187,390	$155,100	
5	$ 317,270	$321,300	$252,400	$227,510	$192,620	
6	$ 389,320	$376,350	$304,540	$271,320	$237,200	
7	$ 452,940	$432,870	$347,480	$315,580	$269,680	
8	$ 522,440	$480,570	$394,640	$355,820	$308,980	
9	$ 572,080	$535,070	$438,550	$398,400	$338,530	
10	$ 635,540	$593,300	$489,470	$443,210	$367,190	
11	$ 687,250	$637,780	$532,670	$479,770	$390,910	
12	$ 758,450	$685,960	$585,220	$524,580	$411,030	

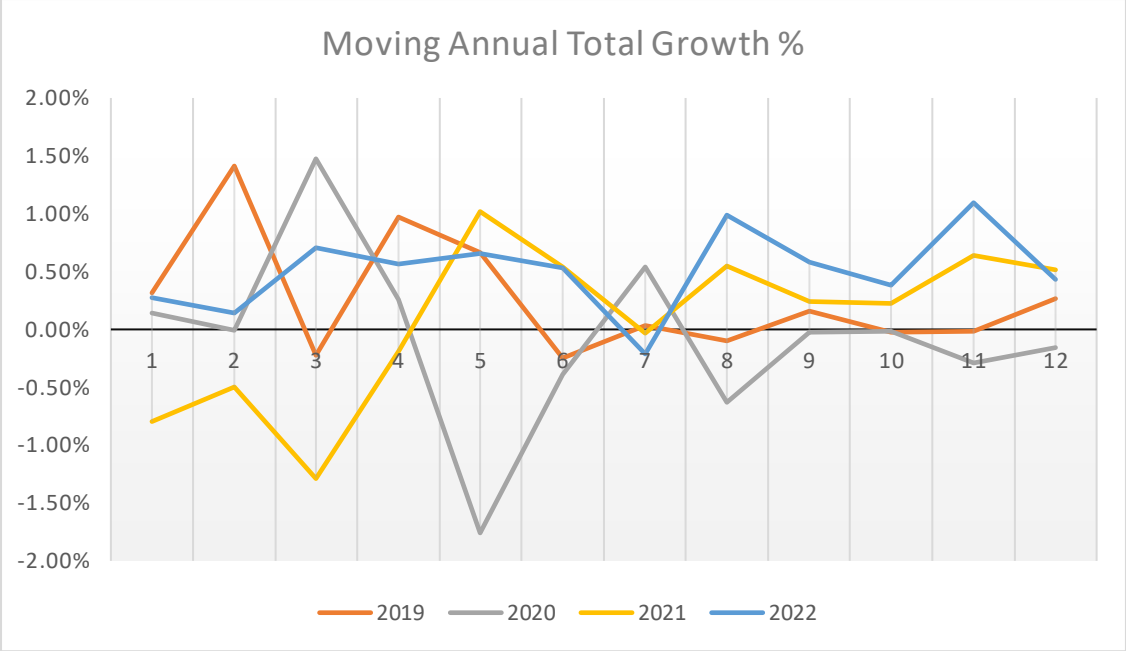
Year 2018 is a good year for redrose and the central region, as it recorded $758,450 value of sales and the lowest sales of all were in 2022.

4. **Moving annual total KPI (Total category value)**: Total of last 12 months updated every month.

Sum of Values MAT	**Column Year**					
Row Month	**2018**	**2019**	**2020**	**2021**	**2022**	**2023**
1	$ 7,242,671	$85,926,035	$88,548,354	$86,934,172	$88,664,049	$95,524,798
2	$ 13,185,173	$87,139,541	$88,541,939	$86,505,007	$88,786,533	$97,139,601
3	$ 20,826,602	$86,941,455	$89,847,515	$85,389,748	$89,414,092	$98,452,906
4	$ 27,291,762	$87,785,569	$90,076,922	$85,229,042	$89,921,857	
5	$ 34,615,539	$88,367,790	$88,492,683	$86,097,118	$90,509,390	
6	$ 42,405,581	$88,149,071	$88,149,132	$86,559,349	$90,987,886	
7	$ 49,733,610	$88,179,005	$88,622,999	$86,528,556	$90,794,731	
8	$ 56,766,646	$88,091,317	$88,067,673	$87,000,644	$91,693,132	
9	$ 63,731,081	$88,227,352	$88,042,637	$87,211,520	$92,222,616	
10	$ 71,458,475	$88,205,712	$88,027,225	$87,405,010	$92,571,615	
11	$ 78,491,683	$88,192,450	$87,771,886	$87,966,225	$93,585,103	
12	$ 85,654,436	$88,424,812	$87,635,108	$88,418,330	$93,991,727	

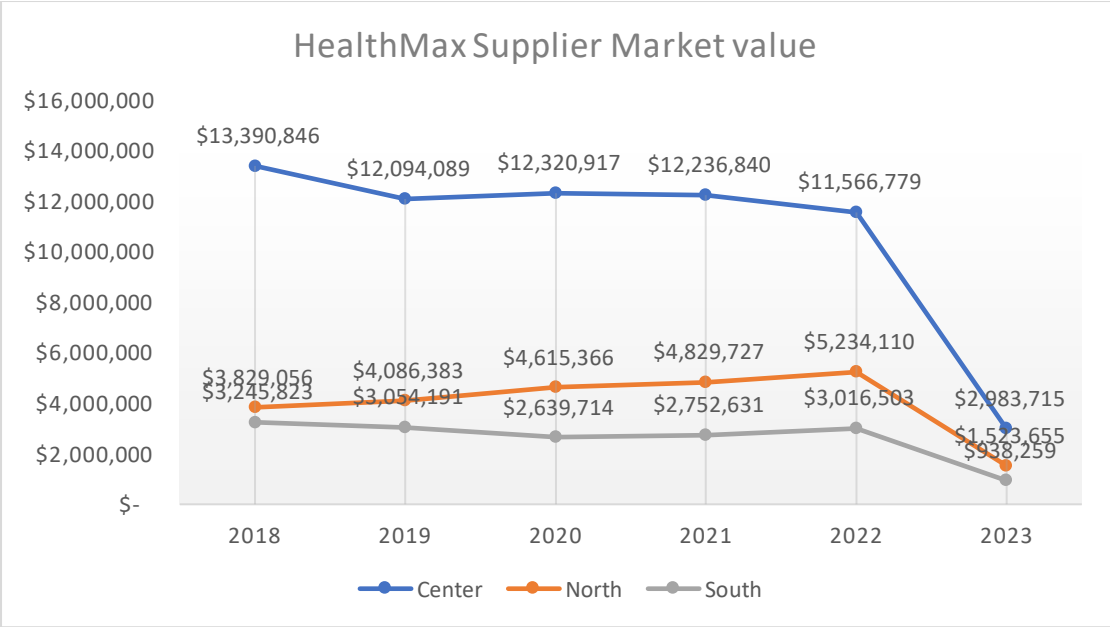
```
=K2+SUMIFS(H:H,D:D,D2,E:E,E2,F:F,F2-1,G:G,">"&G2)
```

A new column is calculated for the Values MAT using the above formula, and from the pivot table data, a line chart is created showing the MAT value growth percentage. Ignored 2018 and 2023 to avoid discrepancies.



MAT growth % is calculated by (Current Period MAT-Previous Period MAT)/Previous Period MAT. Year 2022 is showing the best results in MAT value growth % from its previous 12 months.

5. **Healthmax market value region by region**: Created a pivot table with slicers healthmax-supplier and brand Starbust in place.



For the brand Starbust, the value was very good in the centre region and low value in the south region. Ignore the dip in 2023 and the data is only until March 2023.

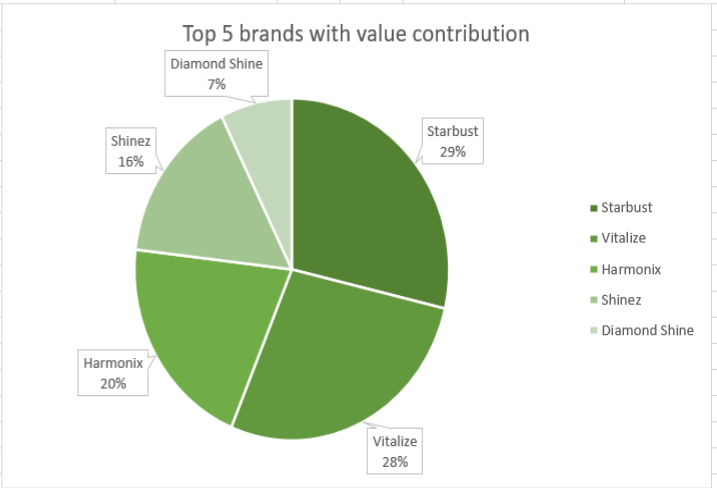
Supplier	HealthMax		
Brand	Starbust		
Sum of Values Month	**Column Labels**		
Row Labels	**Center**	**North**	**South**
2018	$ 13,390,846	$3,829,056	$3,245,823
2019	$ 12,094,089	$4,086,383	$3,054,191
2020	$ 12,320,917	$4,615,366	$2,639,714
2021	$ 12,236,840	$4,829,727	$2,752,631
2022	$ 11,566,779	$5,234,110	$3,016,503
2023	$ 2,983,715	$1,523,655	$ 938,259

Net Revenue Management:

A. **Brand Portfolio Pricing**

1. **Net Sales Contribution %**:

Row Labels	Sum of Values Month	Value Contribution
Starbust	$ 104,358,604.0	28.68%
Vitalize	$ 100,469,075.0	27.61%
Harmonix	$ 74,831,060.0	20.56%
Shinez	$ 57,451,214.0	15.79%
Diamond Shine	$ 26,771,747.7	7.36%
Grand Total	**$ 363,881,700.7**	**100.00%**



These are the top 5 brands that are contributing more to the Net Sales in this industry. Healthmax's brands Starbust and Shinez are with 29% & 16% respectively. The other brands are Vitalise with 28%, Harmonix with 20%, and Diamond Shine 7%.

2. **Net Sales 2022:** Internal sales data is given, where the volume sold in 2022, the net price, the retail price, and the cost of goods and services are given for the healthmax brands Starbust & Shinez. Different retail and net prices have already been set in different pack sizes: 100ml, 125ml, 150ml, 200ml.

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Brand	Product	Pack Size (ml)	ProductID	Retail Price	Net Price	COGS	Volume 2022
Starbust	Ultra Soft	100	Starbust Ultra Soft 100ml	$ 5.00	$ 3.10	$ 0.90	1,156,348
Starbust	Extra Shiny	100	Starbust Extra Shiny 100ml	$ 5.00	$ 2.90	$ 0.95	693,809
Starbust	Strong Hair	100	Starbust Strong Hair 100ml	$ 5.00	$ 2.80	$ 1.00	346,904
Starbust	Ultra Soft	150	Starbust Ultra Soft 150ml	$ 7.00	$ 3.50	$ 1.20	693,809
Starbust	Strong Hair	150	Starbust Strong Hair 150ml	$ 7.25	$ 3.70	$ 1.30	231,270
Starbust	Ultra Soft	200	Starbust Ultra Soft 200ml	$ 9.00	$ 5.00	$ 1.50	289,087
Shinez	Repair	100	Shinez Repair 100ml	$ 6.00	$ 3.50	$ 1.15	1,089,532
Shinez	Repair	125	Shinez Repair 125ml	$ 7.00	$ 4.25	$ 1.30	466,942
Shinez	Repair	150	Shinez Repair 150ml	$ 8.00	$ 5.10	$ 1.50	389,118
Shinez	Repair	200	Shinez Repair 200ml	$ 10.00	$ 6.00	$ 1.70	233,471

Net sales 2022: Volume 2022 * Net Price

Gross Profit/Unit: Net price – cost of goods and services

Gross Profit/Volume: Gross Profit/unit * Volume 2022

Net COGS 2022: COGS * Volume 2022

Gross Margin%: (Net Sales 2022 – Net COGS 2022)/Net Sales 2022

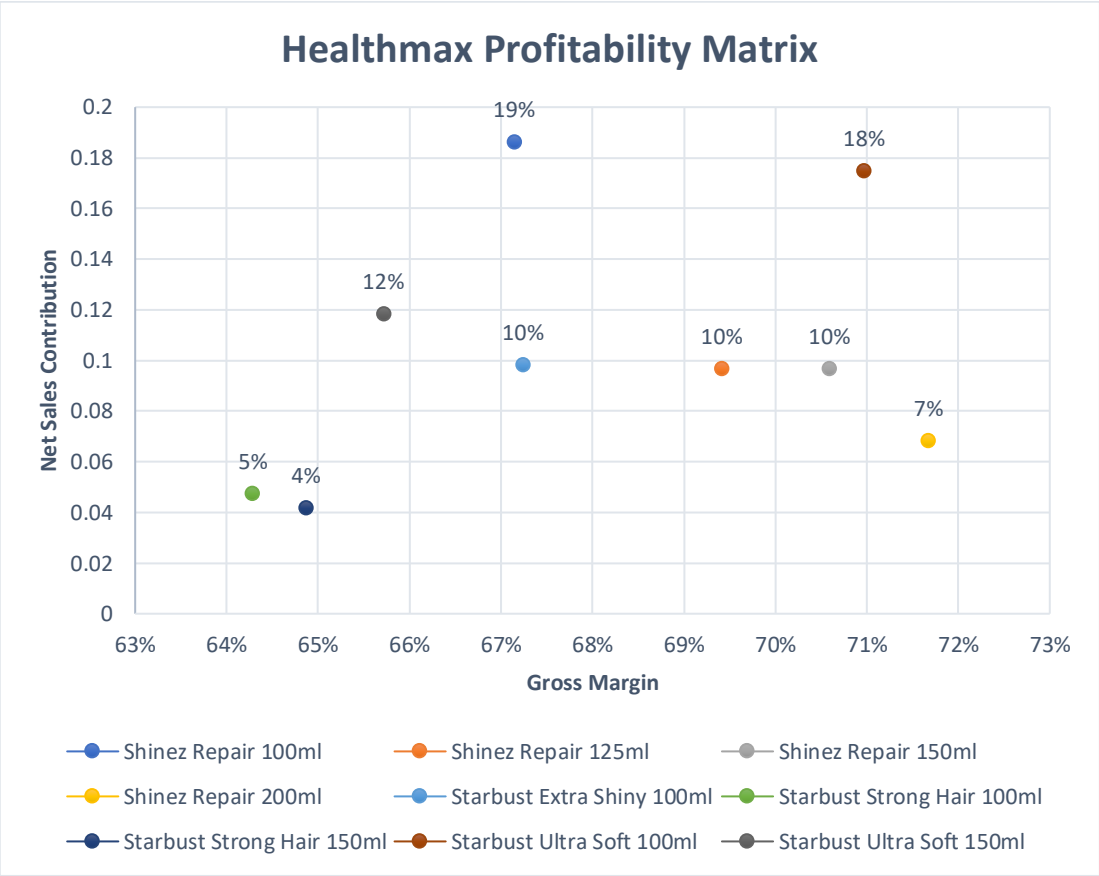
Net Sales Contribution 2022%: Net Sales/Total Net Sales

The above metrics are calculated from the given data.

Net Sales 2022	Gross Profit/Unit	Gross Profit/volum	Net COGS 2022	Gross Margin %	Net Sales Contribution %
$ 3,584,679	$ 2.20	$ 2,543,966	$ 1,040,713	71%	18%
$ 2,012,046	$ 1.95	$ 1,352,928	$ 659,119	67%	10%
$ 971,331	$ 1.80	$ 624,427	$ 346,904	64%	5%
$ 2,428,332	$ 2.30	$ 1,595,761	$ 832,571	66%	12%
$ 855,699	$ 2.40	$ 555,048	$ 300,651	65%	4%
$ 1,445,435	$ 3.50	$ 1,011,805	$ 433,631	70%	7%
$ 3,813,362	$ 2.35	$ 2,560,400	$ 1,252,962	67%	19%
$ 1,984,504	$ 2.95	$ 1,377,479	$ 607,025	69%	10%
$ 1,984,502	$ 3.60	$ 1,400,825	$ 583,677	71%	10%
$ 1,400,826	$ 4.30	$ 1,003,925	$ 396,901	72%	7%

Total Volume 2022	5,590,290
Total Net Sales 2022	$ 20,480,715
Total Gross Profit/Volume	$ 14,026,563
Total COGS 2022	$ 6,454,152
Total Gross Margin %	68%

3. **Healthmax profitability matrix**: Using the gross margin and net sales contribution, a scatter plot is created.



Product Shinez repair 100ml has the highest Net Sales Contribution of 19% with a Gross margin of around 67%.

B. **Mix Management**

1. **New Category Opportunity:** A chance for a company to enter a product category it does not currently sell into to grow revenue and profit. It is found that the most profitable subcategory is organic. The number of units sold in every year until 2022 is shown in the pivot table below.

Sum of Units Month	Column Labels				
Row Labels	2018	2019	2020	2021	2022
Anti-dandruff		2.83%	3.56%	-1.85%	6.70%
Color-safe		-7.99%	-15.04%	-10.78%	-7.12%
Moisturizing		9.43%	-5.31%	4.16%	10.38%
Organic		73.63%	23.07%	43.15%	22.86%
Volumizing		-6.76%	1.26%	-1.18%	-4.65%

Until March 2023, the total units sold were 850749 only in the organic subcategory with a value of $6,837,714. Below, I calculated the units and value if they are increased by reasonable estimates 10% & 12% in the year 2024.

	Units	Value
In yr 24, if the organic category units and value increased by 10%	935824	$ 7,521,485
In yr 24, if the organic category units and value increased by 12%	952839	$ 7,658,239

2. **New product launch**: Healthmax is going to introduce new products called HerbEssentials & Herbashine, possibly organic products with an estimated unit market share of 10% and 12%, respectively. This dataset is called new_product_launch.

Product	Estimated Unit Market Share	Estimated Units Sold 2024	Net Price	COGS
HerbEssentials	10%	935813	$ 3.50	$ 1.35
Herbashine	12%	952828	$ 2.75	$ 0.90

Net Sales, Gross profit per unit, Gross profit per product, Gross Profit Margin, and Net Sales Contribution % are calculated.

Net Sales	Gross Profit per unit	Gross Profit per product	Gross Profit Margin	Net Sales Contribution %
$ 3,275,346	$ 2.15	$ 2,011,998	61.43%	56%
$ 2,620,277	$ 1.85	$ 1,762,732	67.27%	44%

Optimising Net Revenue: Optimising net revenue is a data-driven approach focused on maximising a company's profitability, rather than just sales volume by aligning pricing, promotion, and forecasting.

1. **Pack Price Architecture**: Designing the right price relationship between different pack sizes of the same product. A good PPA protects margins, encourages trade-up, and makes pricing logical.

   The existing pack price architecture is partially logical, but it is inconsistent. While some show expected price per unit decline with increasing pack size, others violate the trade-up principle by pricing larger packs at a higher unit cost. So, a new pricing is recommended.

   For the brands Shinez and Starbust, based on the product and pack size, a new retail price and new price per ml are calculated.

Brand	Product	Pack Size (ml)	Retail Price	Price Per each ml	New Price per each ml	Target price increase %	New Price per ml	New Retail Price
Shinez	Repair	100	$ 6.00	$ 0.0500	$ 0.0500	15.0%	$ 0.058	$ 5.75
Shinez	Repair	125	$ 7.00	$ 0.0500	$ 0.0480	16.5%	$ 0.056	$ 6.99
Shinez	Repair	150	$ 8.00	$ 0.0500	$ 0.0470	8.0%	$ 0.051	$ 7.61
Shinez	Repair	200	$ 10.00	$ 0.0467	$ 0.0450	11.0%	$ 0.050	$ 9.99
Starbust	Ultra Soft	100	$ 5.00	$ 0.0483	$ 0.0550	15.0%	$ 0.063	$ 6.33
Starbust	Extra Shiny	100	$ 5.00	$ 0.0450	$ 0.0550	15.0%	$ 0.063	$ 6.33
Starbust	Strong Hair	100	$ 5.00	$ 0.0600	$ 0.0550	15.0%	$ 0.063	$ 6.33
Starbust	Ultra Soft	150	$ 7.00	$ 0.0560	$ 0.0500	8.0%	$ 0.054	$ 8.10
Starbust	Strong Hair	150	$ 7.25	$ 0.0533	$ 0.0500	8.0%	$ 0.054	$ 8.10
Starbust	Ultra Soft	200	$ 9.00	$ 0.0500	$ 0.0470	11.0%	$ 0.052	$ 10.43

Pack size (ml)	Target price increase
100	15.0%
125	16.5%
150	8.0%
200	11.0%

For the brand Shinez, the price per ml is identical for 100, 125, and 150 ml. There is no financial incentive for the consumers.

The new prices ensure a logical trade-up incentive, protect margin and also align with the PPA principles. Also, there is a price increase from the business point of view.

2. **Promotion Analysis**: In the year 2022, there were 3 different promotion events in April, August, and November. This part is the analysis of that. The data can be found in the promotion analysis sheet.

Promotion	Mechanism	Brand	Year	Month	Value Sales	Baseline Sales	Uplift Amount	Cost	ROI
1	Buy 2, get 1 free	Shinez	2022	4	$ 1,445,626	$ 840,836	$ 604,790	$620,000	-2%
2	Buy 2, get 20% off	Shinez	2022	8	$ 1,198,708	$ 840,836	$ 357,872	$320,000	12%
3	2nd at 50% off	Shinez	2022	11	$ 1,413,646	$ 840,836	$ 572,810	$540,000	6%

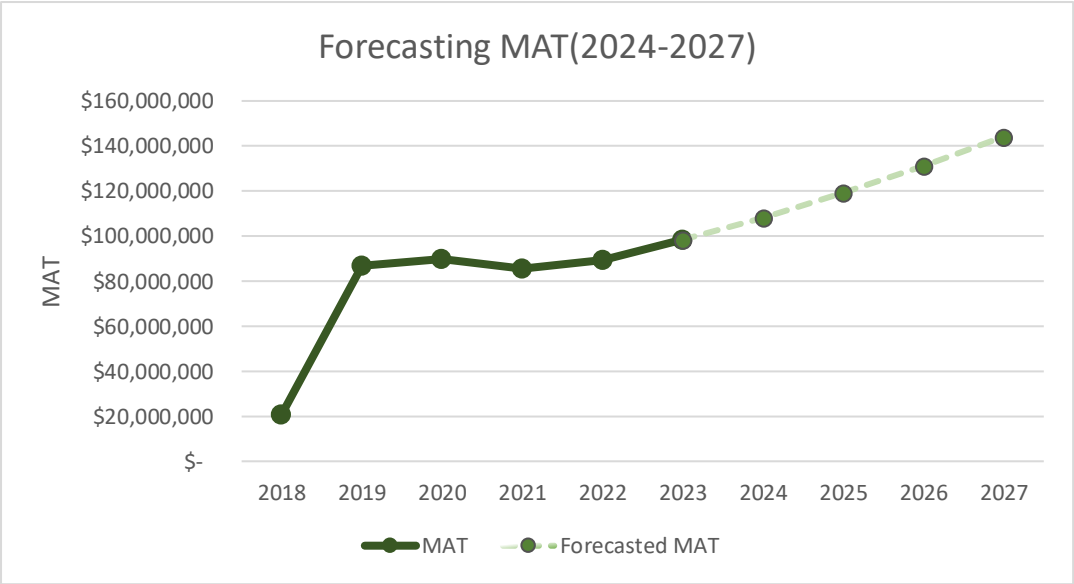
The baseline sales, uplift amount and ROI% is calculated from the pivot table Shinez Value 2022. It can be seen that the promotion event, which was held in August with "Buy 2, get 20% off" have the highest ROI of 12%.

3. **Forecasting MAT**:

The final part of optimising the net revenue is forecasting MAT from 2024 to 2027. Before forecasting the MAT, the current 2023 MAT until March is taken, which is $98,452,906 and the following additions are made

Current Value	$ 98,452,906
organic Growth-4%	$ 3,938,116
Price increase-3%	$ 2,953,587
Mix product management-2%	$ 1,969,058
New Organic product	$ 1,000,000
Total Forecast Amount	$ 108,313,667
Total Growth %	10.02%

The 10% annual MAT growth assumption reflects a combination of organic market expansion and incremental gains from pack price architecture optimisation, improved trade, and new product launch within the sub-category.



The above line chart suggests that the FMCG industry will have positive results in the future. The forecast reflects the combined effect of natural market growth, new pricing, PPA, and promotion.

Conclusion:

The Net Revenue Management analysis demonstrates how strategic optimisation across pricing, product mix, trade investment, pack price architecture, and promotional effectiveness can drive sustainable and profitable revenue growth.

Through Brand Portfolio Pricing and Pack Price Architecture, a more logical and strategic price structure was developed to support consumer trade-up and protect margins. The evaluation of a new product opportunity helped strengthen the overall portfolio by increasing the share of higher-margin products.

By combining these initiatives with organic market growth, a structured revenue forecast was developed, projecting continued MAT growth over the coming years. Overall, this analysis highlights how integrated NRM strategies can enhance both revenue and profitability, providing a strong foundation for long-term business performance.

CHEERS!